

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech.(EC) Examination

November 2013

EC 201 Digital Electronics & Logic Design

Date: 21.11.2013, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

- Q - 1 (a)** Convert the following numbers to decimal: (1) $(12121)_3$ (2) $(50)_7$ (3) $(8.3)_9$ (4) $(198)_{12}$ [04]
- (b)** Find the 10's Complement of $(935)_{11}$. [01]
- (c)** Define following terms: (1) Power Dissipation (2) Propagation Delay [02]
- Q - 2 (a)** Convert the following functions in sum of minterms and product of maxterms. [07]
- (1) $F(A,B,C) = (A'+B)(B'+C)$
- (2) $F(A,B,C) = (AB+C)(B+AC)$
- (3) $F(A,B,C) = 1$
- (b)** Simplify the Boolean function F using don't care conditions d, in (1) sum of products [07]
- (2) product of sums.
- $F(A,B,C,D) = \sum (1,3,7,11,15)$ and $d(A,B,C,D) = \sum (0,2,5)$

OR

- Q - 2 (a)** Simplify the following Boolean function by means of the tabulation method [07]
- $F(A,B,C,D) = \sum (0,1,2,8,10,11,14,15)$
- (b)** Design a combinational circuit that generates the 9's complement of a BCD digit. [07]
- Q - 3 (a)** Obtain NAND logic diagram of a following boolean function: [04]
- $C = XY + YZ + XZ$
- (b)** Explain Magnitude Comparator in detail. [05]
- (c)** Implement the following function with a multiplexer: [05]
- $F(A,B,C,D) = \sum (0,2,4,6,8,9,11,14,15)$

OR

- Q - 3 (a)** Draw the K-Map for a Four variable (1) Exclusive-OR function and (2) Equivalence Function. [04]
- (b)** Design a combinational circuit using a ROM. The circuit accepts a 3-bit number and generates an output binary number equal to the square of the input number. [05]

- (c) A combinational Circuit defined by the functions. [05]
 $F_1 = \sum(3,5,6,7)$ and $F_2 = \sum(0,2,4,7)$
 Implement the circuit with Programmable Logic Array(PLA).

SECTION – II

- Q - 4 (a)** Given the Boolean function: [04]
 $F = AB + A'B' + B'C$
 (1) Implement it with only OR and NOT gates.
 (2) Implement it with only AND and NOT gates.
- (b)** Show the Logic Diagram, Characteristic Table and Characteristic Equation of clocked RS flip-flop. [03]
- Q - 5 (a)** Explain Operation of the D-type edge-triggered flip-flop. [07]
- (b)** Design a counter that counts the decimal digits according to 2,4,2,1 code. Use T flip-flops. [07]

OR

- Q - 5 (a)** Design a sequential circuit with four flip-flops A, B, C and D. The next states of B, C and D are equal to the present states of A, B and C respectively. The next state of A is equal to the exclusive-OR of the present states of C and D. [07]
- (b)** Design a counter with the following sequence: 0, 1, 2, 3, 4 and repeat. Use JK flip-flops. [07]
- Q - 6 (a)** Define Register. Explain 4-bit register with parallel load. [07]
- (b)** Explain BCD Ripple Counter with necessary figure. [07]

OR

- Q - 6 (a)** Design 4 bit binary Up-Down Counter with necessary figure. [07]
- (b)** Explain Memory Unit with necessary Block Diagram.. [07]
